## Task 1: Explore and understand the dataset

*1. What is the unique number of stations in the dataset?*

*2. How many Citibikes is in the dataset?*

*3. What is the most popular start station and end station?*

*4. What is the gender distribution of the Citibike users?*

*5. What is the usertype distribution of the Citibike users?*

*6. What is the trip duration distribution of the Citibike trips?*

*7. What is the most popular Citibike trip?*

*8. Was there new bike introduced or removed at any point in time? What makes you think it did or didn't? What about the citibike stations?*

**For the above tasks I have used 2 datasets (citibike_stations and citibike_trips from January 2015).**

We can drop the the columns that are not needed for resolving the above tasks.

Columns that can be dropped from citibike_stations are:

Short_name, Latitude, Longitude, region_id, rental_methods, eightd_has_key_dispenser, eightd_has available keys.

Columns that can be dropped from citibike_trips_Jan2015:

start_station_longitude, start_station_latitude, end_station_longitude, end_station_latitude, birth_year.

***1.***

**SELECT COUNT(DISTINCT station_id) AS ID FROM citibike_stations**

There are 935 unique stations in the dataset.

***2.***

**SELECT COUNT(DISTINCT bikeid) AS NumberOfBikes FROM citibike_trips_Jan2015**

**SELECT SUM (num_bikes_available) as totalQuantityAVAILABLE FROM citibike_stations**

**SELECT SUM(num_bikes_disabled) as totalQuantityDISABLED FROM citibike_stations**

If we use the data from January 2015 our result will be 4271 , however if we use the data from citibike_stations which isn't limited to January of 2015 we will see that there are in total

12683 available bikes and 360 disabled bikes which makes a total of 13043 bikes.

***3.***

**SELECT start_station_id,COUNT(*) as PopularStart FROM citibike_trips_Jan2015 GROUP BY start_station_id ORDER BY PopularStart DESC**

**select end_station_id,COUNT(*) as PopularEnd FROM citibike_trips_Jan2015 GROUP BY end_station_id ORDER BY PopularEnd DESC**

Most popular start station is the station by the ID 521 – 4157 starts.

Most Popular end station is the station by the ID 521 – 3319 ends.

ID 521 is "8 Ave & W 31 St" station.

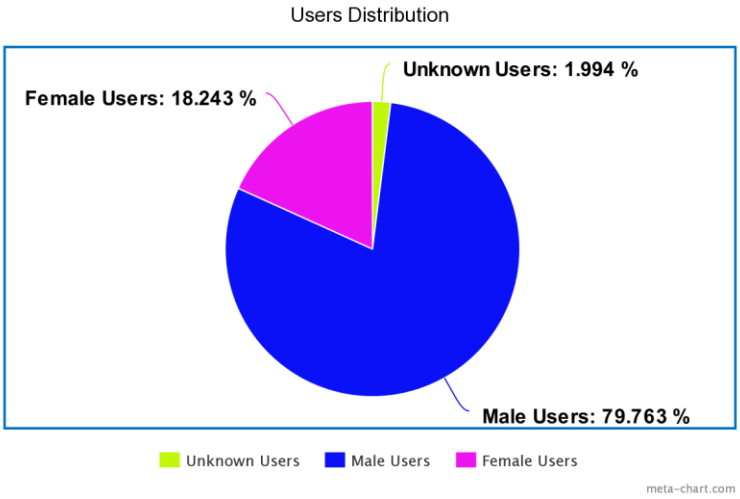
***4.***

**SELECT gender,COUNT(*) as Result FROM citibike_trips_Jan2015 GROUP BY Result**

5695 are Unkown users, 227764 are Male users and 52093 are Female users which gives us a total of 285552 users.

The distribution in percentage is:

1.994% for Unknown Users, 79.763% for Male Users, 18.243% For Female Users.



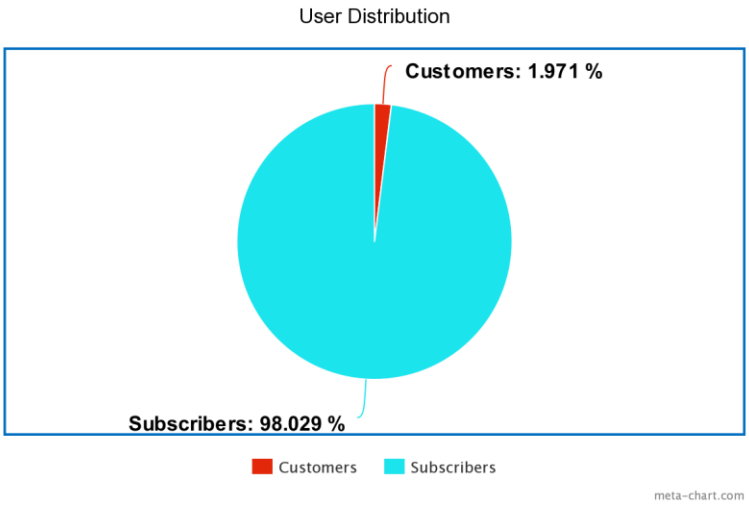

***5.***

**SELECT usertype,COUNT(*) as Result from citibike_trips_Jan2015 GROUP BY usertype**

5628 are Customers and 279924 are Subscribers, which gives us a total of 285552 users.

The distribution in percentage is:

1.971% for Customers and 98.029% for Subscribers.




***6.***

**SELECT sum(cnt) FROM (SELECT tripduration,COUNT(*) as cnt FROM citibike_trips_Jan2015 WHERE tripduration <61 GROUP BY tripduration ) as c**

**SELECT sum(cnt) FROM (SELECT tripduration,COUNT(*) as cnt FROM citibike_trips_Jan2015 WHERE tripduration between 61 and 480 GROUP BY tripduration ) as c**

**SELECT sum(cnt) FROM (SELECT tripduration,COUNT(*) as cnt FROM citibike_trips_Jan2015 WHERE tripduration >480 GROUP BY tripduration ) as c**

Total number of trips is 285552.

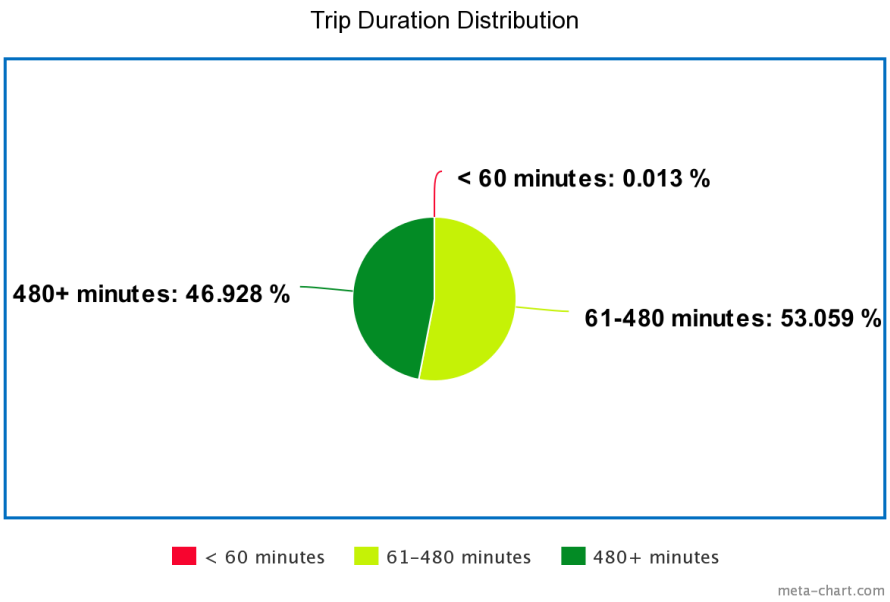
We have created a 3 groups of trips-

0-60min, 61-480min, and 480+ min.

There were 37(0.013%) trips which lasted for less than 61 minutes.

There were 151513(53.059%) trips which lasted between 61 and 480 minutes.

There were 134002(46.928%) trips which lasted for more than 480 minutes.




***7.***

**SELECT CONCAT(start_station_id, '' ,end_station_id) AS Most_Popular_Trip**

**INTO temp123 FROM citibike_trips_Jan2015**

**SELECT COUNT(1) AS Number_Of_Trips, Most_Popular_Trip**

**FROM temp123 GROUP BY Most_Popular_Trip ORDER BY Number_Of_Trips DESC**

**SELECT * FROM citibike_trips_Jan2015 WHERE start_station_id like 318**

**SELECT * FROM citibike_trips_Jan2015 WHERE end_station_id like 477**

For this task we have merged 2 columns(start_station_id and end_station_id) and we have placed the new column into temporary table called temp123.

This way we have gained a Unique ID for each trip - for example if the start_station_id was 123 and the end_station_id was 456, the ID for that trip would have been 123456.

By doing the above query, we have gained an insight that the most popular trip is 318477 with the total amount of 303 trips.

Start station for most popular trip is E 43 St & Vanderbilt Ave – Station ID 318,

and the end station for the most popular trip is W 41 St & 8 Ave – Station ID 477.

## *8.*

We don’t know for sure if there were new bikes introduced at any point in time, as we don’t have unique bike ID’s in bike_stations table, which would have helped us to gain an insight into unique number of bikes at the current stations.

As for the the new citibike stations, we are running into a problem which got triggered by the lack of data.

From the given table we can only see the last report issued(which is mostly from late January 2020, with the exception of few from 1970), while we are lacking the data for the first report issued. If we had a first report issued, we could have drawn the line between that time frame and between the start-stop time from citibike_trips table(as all of the trips from that table are from January 2015).